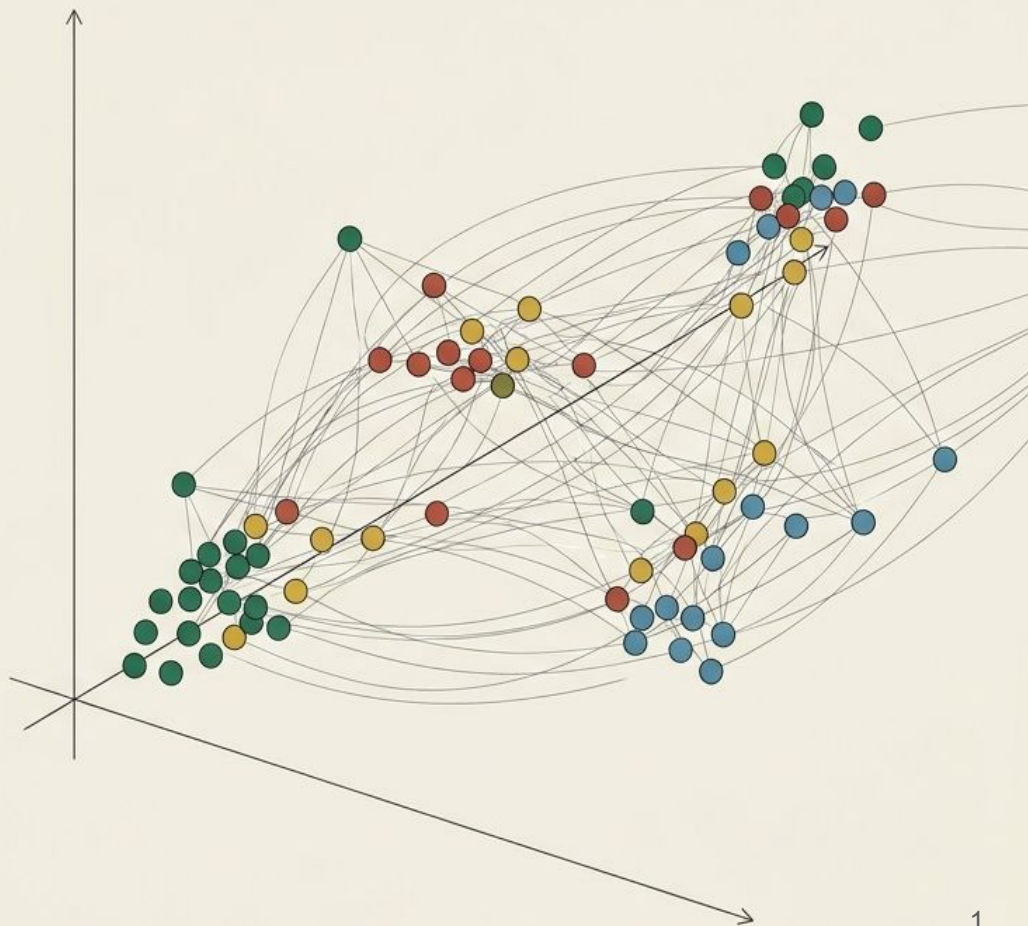


Integrating Cognition into Migration Models

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Existing Migration Studies vs. Empirical Reality

Traditional Models

Portraying human mobility through two contrasting lenses:

- Homogenous economic actors: Utility-maximizing individuals.
- Passive victims: Driven by external forces like conflict or disasters.

Empirical Reality

- Models fail to explain why **identical structural positions** lead to different decisions.
- Individuals may feel "trapped" despite a desire to move.
- Others stay in home communities despite economic incentives.



Moving Beyond Binaries

Migration is a multifaceted reality embedded in structural inequalities, cognitive landscapes, and social contexts.

Psychological factors such as attitudes,
behavioral control, or social norms
are seldom explicitly modeled.
— Klabunde and Willekens (2016)

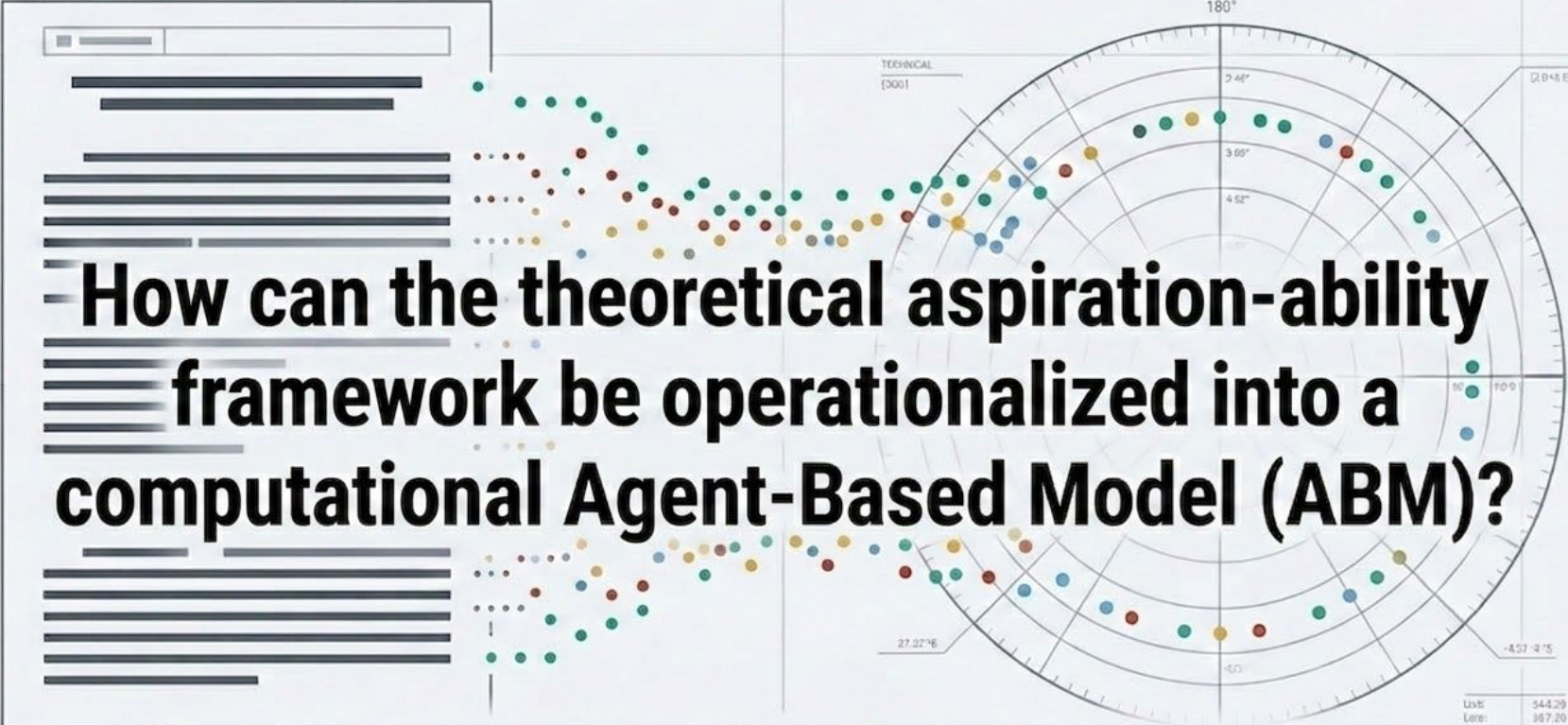
**Structural
Forces &
Economics**

**Spatial
Movement &
Migration**



Carling & Schewel's (2018) Aspiration-Ability Framework:

Distinguishing the internal desire to migrate from the perceived external ability to do so.



How can the theoretical aspiration-ability framework be operationalized into a computational Agent-Based Model (ABM)?

To introduce human agency and cognitive complexity into computational migration research, revealing the hidden, heterogeneous dynamics of global movement.

Carling & Schewel 2018 Aspiration-Ability Framework

Perceived Ability (Low to High)

Content

Low aspiration, variable ability. Possess means but choose to stay.



Opportunity Seekers

High aspiration, high ability. Voluntary migrants with requisite resources.



Acquiescent

Low aspiration, low ability. Lack both desire and means.



Trapped

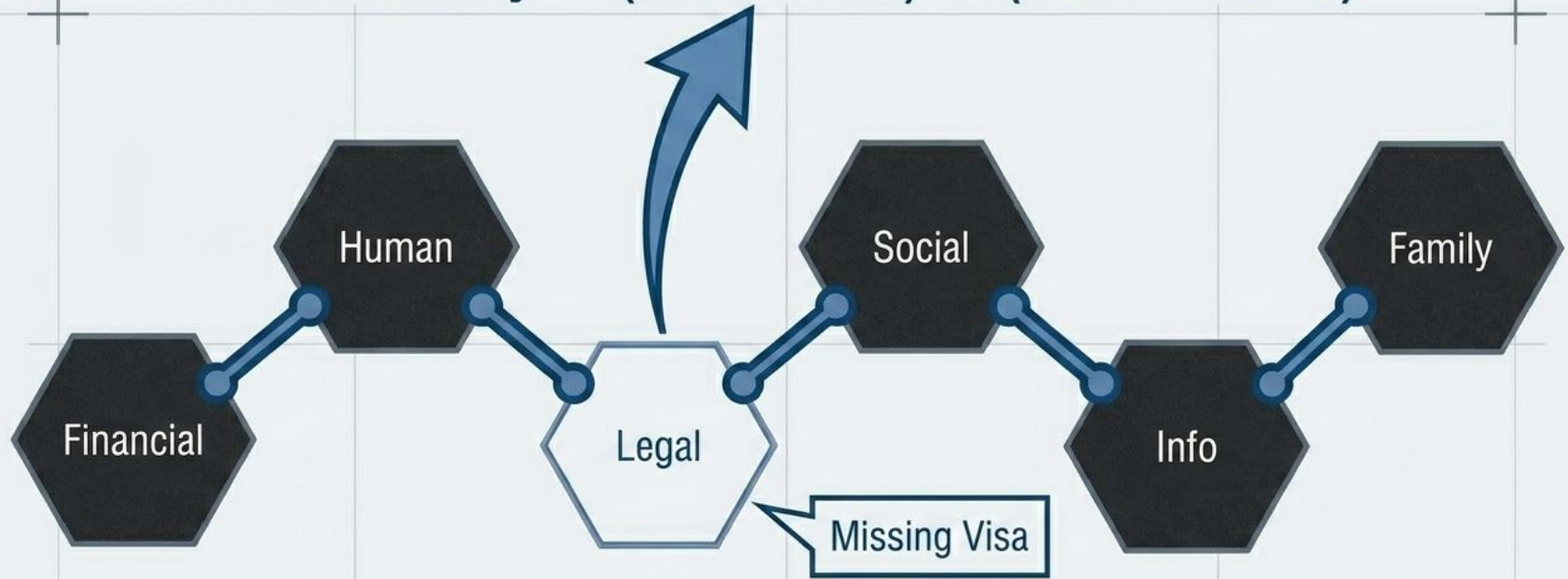
High aspiration, low ability. Desiring movement but constrained by barriers.



Cognition is operationalized through agents' perceived ability and their socially mediated aspiration formation.

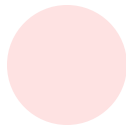
Aspiration to Migrate (Low to High)

$$\text{Total Ability} = (0.2 \times \text{min}) + (0.8 \times \text{mean})$$



The Structural Reality: Modeling Ability. A compensatory system where significant bottlenecks—like a missing visa—heavily drag down the average, stranding an agent in the Low Ability threshold.

The Hierarchy of Aspiration: Safety-First Logic



Condition 1: Safety Threshold Not Met

Aspiration is 100% safety-driven. All other factors are muted until immediate survival is secured.



Condition 2: Safety Requirements Satisfied

The "Maslow Bottleneck" opens, allowing higher-level aspirations to form via a weighted blend.

70%

Internal Desires

Economic, Family, Adventure

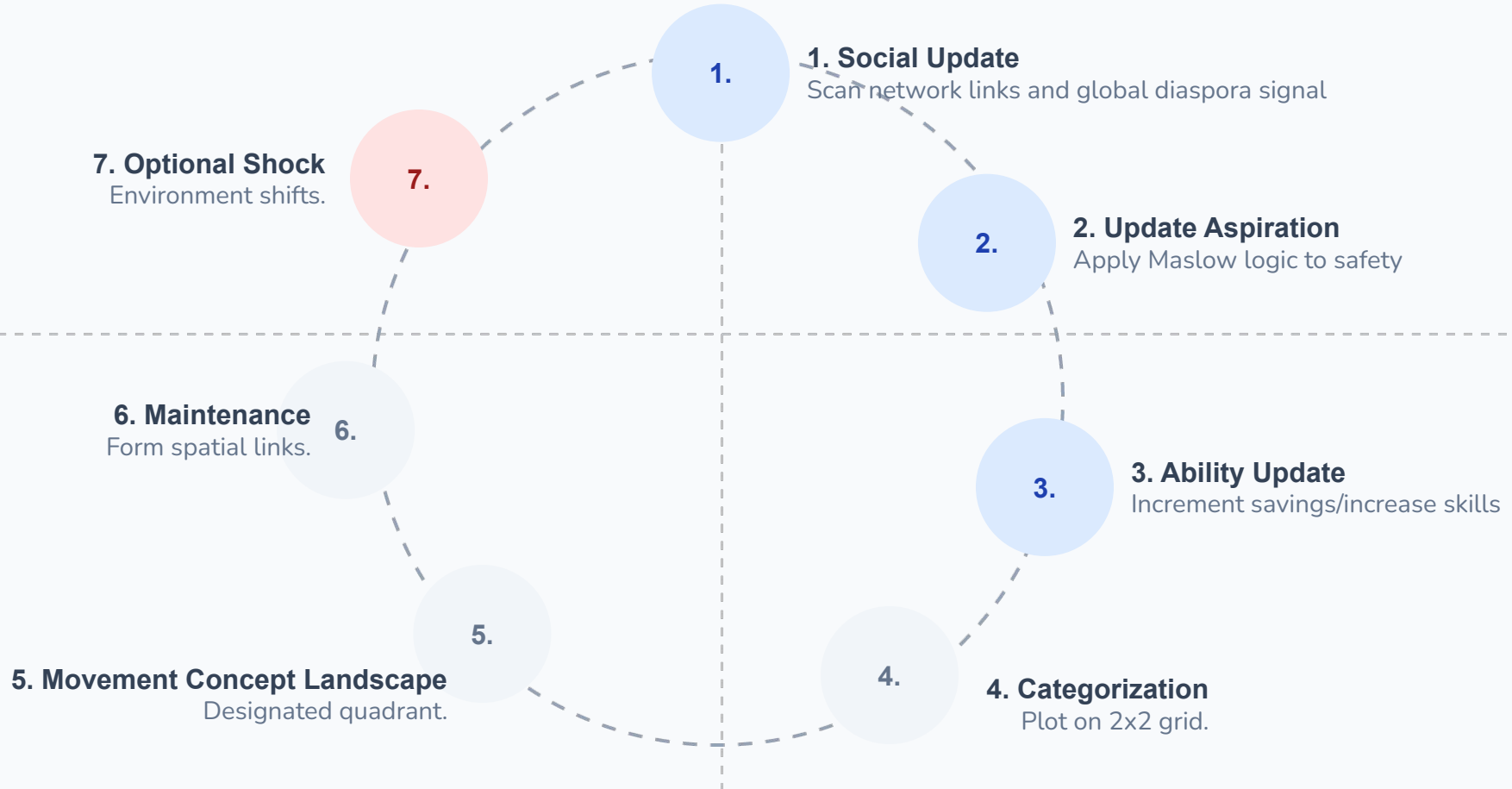
30%

Social Influence

Neighbors & Global Diaspora

Model Logic: Aspiration = If(Safety < Threshold, 1.0 * Safety, 0.7 * Internal + 0.3 * Social)

The Cognitive Engine Cycle



Simulation Scenarios & Migration Phenomena

Run 1: Culture of Migration

Primary Driver
Social Networks/Human Capital

Aspiration State
Emerging/Evolving

Ability State
Steady Growth

Gap (Asp. vs Ability)
Narrow & Manageable

Run 2: Economic Downturn

Primary Driver
External Disruption

Aspiration State
Spiked/Inflated

Ability State
Sharp Decline

Gap (Asp. vs Ability)
Wide & Frustrated

Run 3: High Aspiration

Primary Driver
Capability Accumulation

Aspiration State
Constant (High)

Ability State
Steady Growth

Gap (Asp. vs Ability)
Closing over time

Model Scale: 200 Individual Agents

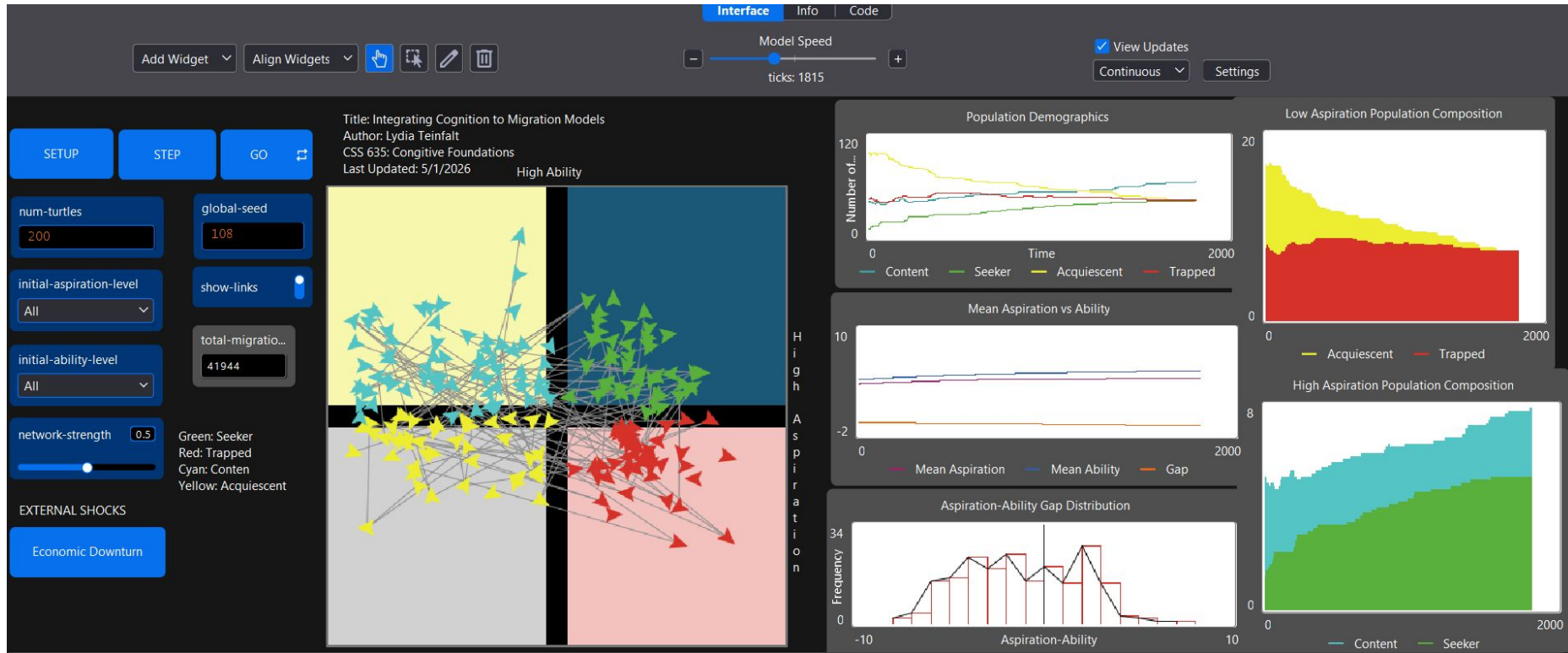
Duration: 1825 Ticks (1 tick = 1 day)

Agents initialized with aspiration/ability (1-10). Ability grows via human capital simulation (learning/savings).

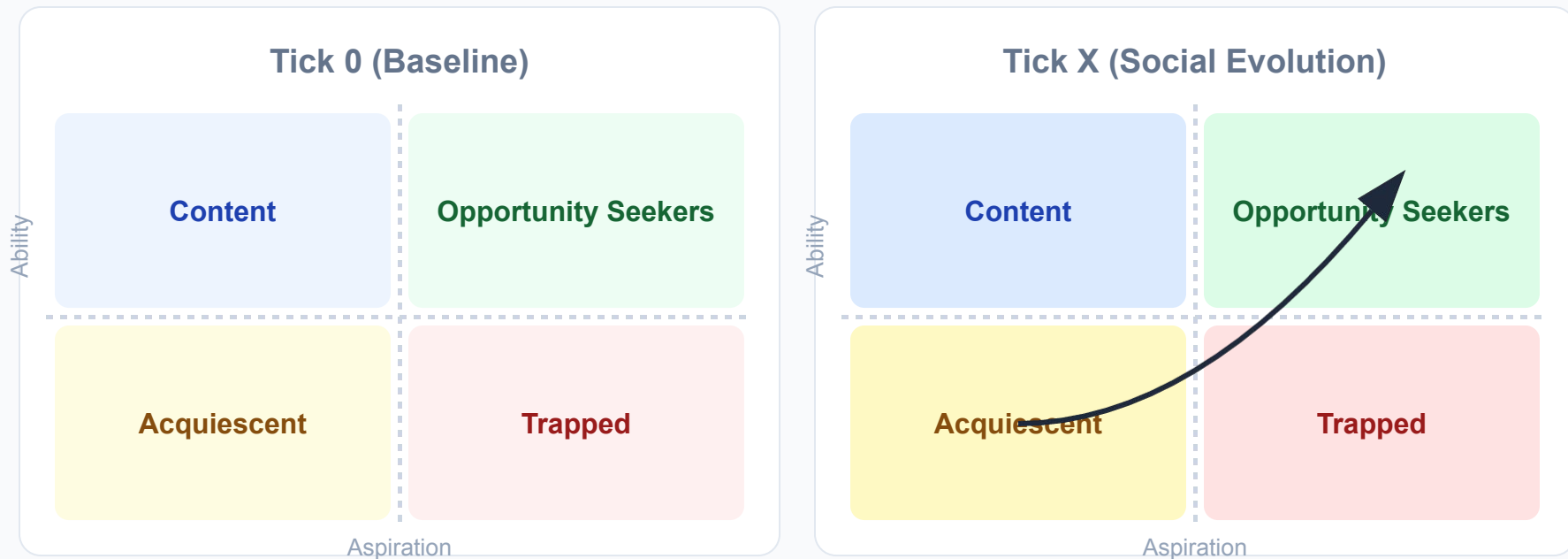
Aspiration increases through randomly assigned network connections.

Diaspora signal strength is calculated based on the density of opportunity seekers within the network.

NetLogo 7.03 Model

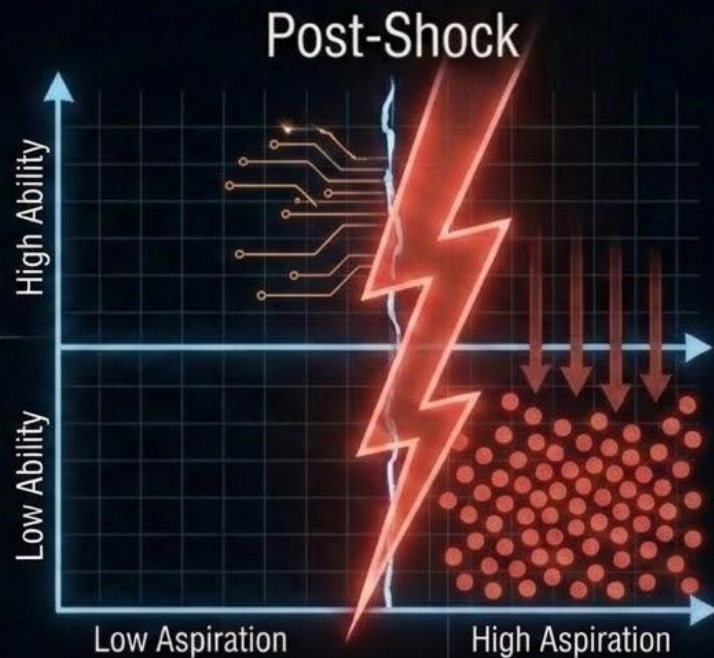
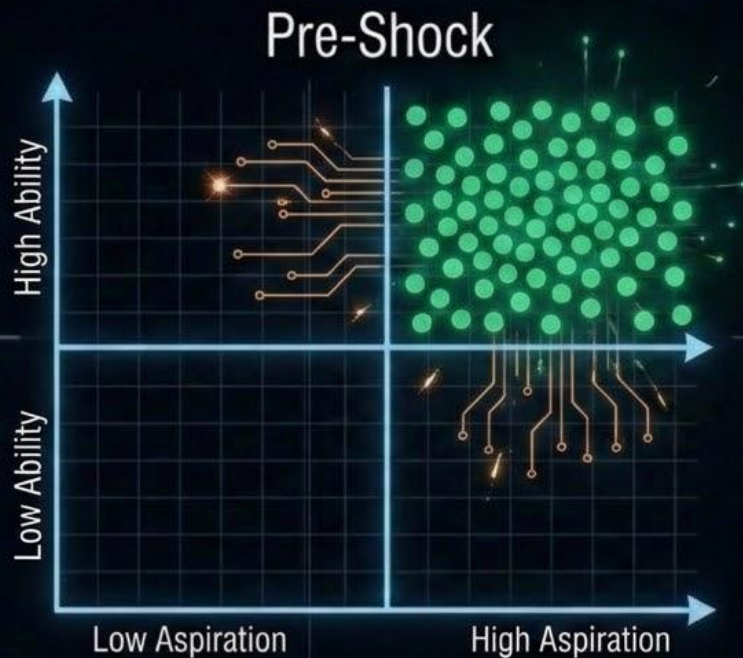


Baseline Simulation: The Culture of Migration



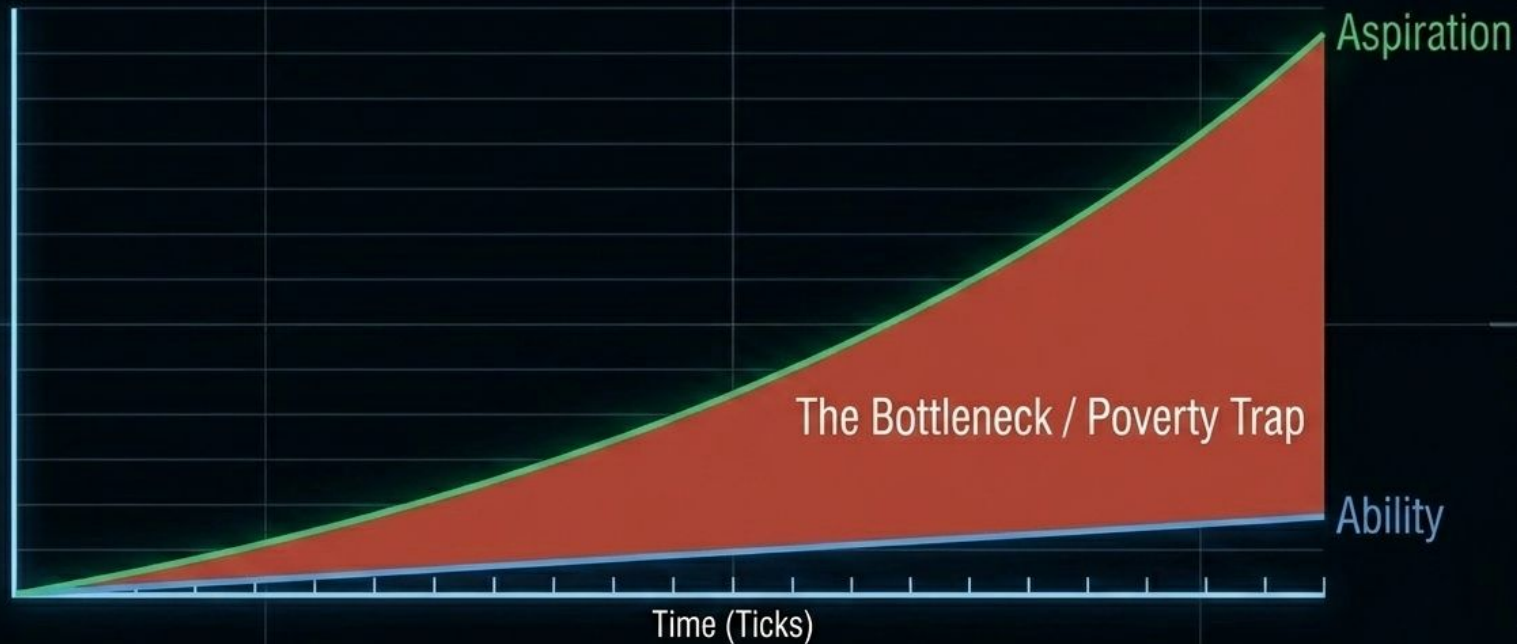
Insight: Natural social evolution. As the diaspora signal grows, the internal drive for those remaining increases. Aspiration is socially constructed and awakened by networks, regardless of immediate economic reality.

Simulation Shock: Creating Cognitive Misalignment



Insight: Systemic failures don't just stop movement—they strip away ability without erasing desire. Opportunity Seekers are instantly transformed into Trapped populations, creating a structural precursor to irregular migration.

Blocked Incremental Actualization: The Poverty Trap



Blocked Incremental Actualization. While ability grows linearly through savings, aspirations compound socially. The inescapable gap creates a permanent cognitive bottleneck for populations in precarious conditions.



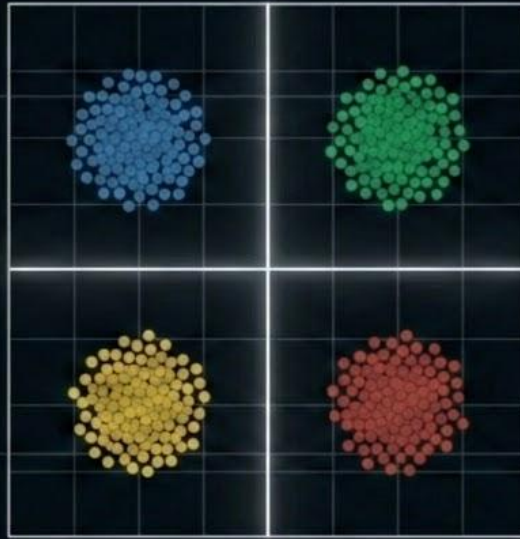
MIGRATION IS NOT A SIMPLE RESPONSE TO POVERTY.

Poverty is frequently a barrier to migration, not a driver. The most dangerous systemic state—the “Cognitive Gap”—occurs when aspirations remain high but structural ability is stripped away, leaving a frustrated population indefinitely trapped.

Scope & Computational Limitations

The International Focus:
Distance, information asymmetries, and hard structural barriers (like visas and language) uniquely magnify the gap between aspiration and ability.

Quantifying the Subjective:
A fundamental limitation of Agent-Based Models is the necessary oversimplification of treating subjective human emotions as rigid Cartesian coordinates.



By integrating cognition, Agent-Based Models cease to be mere economic calculators. They become engines for empathy, capable of simulating the complex psychological state of global migration.

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<https://github.com/lydiateinfalt/Agent-Based-Modeling-Migration-and-Cognition>

Selected images from slides were created with the NotebookLM and Advanced Gemini

Validation Matrix: Theory to Observable ABM Result

Carling & Schewel (2018) Concept

Observable ABM Result

Involuntary Immobility
(High desire blocked by walls)

Density of the Red Cluster

Social Construction of Aspiration
(Desires shaped by networks)

Transition of Yellow nodes
to Red/Green over time

Hierarchical Needs
(Safety precedes migration)

Maslow Bottleneck logic
halting total aspiration

Capability Accumulation
(Migration requires capital
thresholds)

Slow linear ability growth
over time

